

AI and Army Operational Effectiveness

A first-principles analysis for senior Army leadership

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Question	How can Army use AI to increase operational effectiveness?

The Question

How can Army use AI to increase operational effectiveness?
What should Army be able to do better because AI exists?

1. This paper reasons from Army's outputs backwards to the technology. It does not begin from the assumption that Army must adopt AI, and it does not catalogue uses. It asks where AI could materially improve Army's ability to generate and employ land capability, what must remain human, and what that means for Army's people and their training.
2. AI is new within the NZDF. Access is limited to basic tools on constrained networks, and the evidence for military effect is thin and drawn largely from larger armies with different problems. Where this paper asserts, the assertion rests on reasoning from Army's own situation. Where the evidence is insufficient, it says so.

1. The First-Principles Answer

3. Army exists to generate land forces ready for operations, to employ them, and to learn from doing both. Almost everything else Army does supports those three outputs.
4. For a small army the binding constraint on all three is skilled human capacity. There are not enough experienced people, and those there are spend a large share of their time on staff and administrative work that consumes judgement without exercising it. Depth is thin, and adaptation is slow because the people who must adapt are the people already fully committed.
5. AI increases operational effectiveness where it relieves that constraint in ways that convert into capability: faster and better planning; clearer understanding of the force and its environment; more adaptive generation of trained people; faster institutional learning; and the return of experienced people's time to judgement, leadership and training. It does not increase effectiveness where it lets administration be done faster and the hours are absorbed by more administration.

Army should use AI to see itself and its environment more clearly, decide faster with better options, generate and adapt trained people more quickly, and learn from experience at a pace the force can act on, while remaining able to fight when the tools are denied.

6. Three distinctions organise what follows. **Genuine military advantage** changes what Army can do on operations or how fast it can generate capability. **Administrative efficiency** frees time, and becomes advantage only if leaders redirect the time. **Technological novelty** is neither, and is the category most AI proposals fall into.

2. Where AI Creates Military Advantage

7. Five areas pass the test. They are ordered by the strength of the mechanism and by how much of the opportunity Army itself controls.

Planning and staff work

8. The appreciation, course of action development, orders production and the routine staff estimate are language and structure tasks that consume the scarcest people in a headquarters. AI can generate and stress-test options, draft and check products against intent and format, and surface omissions. The mechanism is speed and breadth of option generation, and staff time returned to judgement. The mode is augment: the decision, the acceptance of risk and the intent remain the commander's. The risk is a plausible product that is wrong in a detail no one checks, and a staff that loses the ability to run the process unaided.

Understanding: intelligence, information and the state of the force

9. A small intelligence cell cannot process the open-source, coalition and sensor information now available to it. AI can fuse, summarise, translate and flag change across volumes a person cannot read. Separately, Army's own readiness data on people, training, equipment and health sits in disconnected systems and spreadsheets, and commanders see it late and in part. The mechanism is scale and decision advantage: understanding sooner, from more. For intelligence the mode is augment, and much of the opportunity is enabled by allies; Army's advantage lies in competent consumption and contribution, not in building the system. For readiness the mode is enable, because Army cannot currently do this at all, and the limit is data quality rather than technology.

Generating trained people

10. Training design, content maintenance, scenario and simulation generation, individual practice, feedback and assessment support are the heart of force generation and the point where the training approach and AI meet. AI can make adaptive pathways practical at scale, keep content current as tactics and equipment change, and shift instructor effort from transmission to coaching. The mechanism is readiness, through faster and more adaptive generation of competence. The mode is augment and enable. Section 5 develops this.

Learning and adaptation

11. Army's lessons cycle runs on documents and committees, and is slower than the tempo at which threats, tactics and equipment change. AI can compress the path from observation, after-action review and report to updated training content and doctrine. The mechanism is institutional learning speed, which for a small army is survival, because it cannot afford to relearn. The mode is enable. The uncertainty is whether the organisation will act at the speed the tool allows.

Administrative and organisational load

12. Correspondence, reporting, returns, minutes, records and routine staff process will yield the largest and earliest gains in hours. The mechanism is human capacity, and the mode is automate where consequence is low. This is efficiency, not advantage, unless leaders convert the hours into training, planning and readiness. That conversion is a command decision. Without it the gain is real but not military.
13. Two areas are deliberately not prioritised. Autonomous targeting and weapons are not Army's to develop, are policy and coalition dependent, and are the subject of unresolved legal and ethical questions. Predictive logistics and personnel analytics are real opportunities constrained by data maturity and, for personnel, by bias and trust; they follow the data foundations rather than lead them.

Area	Mechanism	Mode	Consequence of error	Human role
Planning and staff work	Speed, breadth of options, staff time returned to judgement	Augment	Medium to high	Decides, accepts risk, verifies the product
Intelligence and readiness	Scale; understanding sooner, from more	Augment; enable for readiness	High for intelligence; medium for readiness	Assesses sources, judges, decides
Generating trained people	Adaptive competence at scale; readiness	Augment and enable	Low to medium	Coaches, assesses, assures the standard
Learning and adaptation	Speed of institutional learning	Enable	Low to medium	Validates, decides what changes
Administrative load	Human capacity	Automate	Low	Owens what is signed

3. What Should Remain Human

14. The following remain with people, not because AI cannot contribute but because responsibility cannot be delegated to a system:
 - a. The decision to use force, the acceptance of risk, and the statement of intent.
 - b. Assessment of a soldier's readiness, character and leadership, and the coaching relationship that develops them.
 - c. Legal and ethical judgement, including the law of armed conflict and rules of engagement.
 - d. Accountability. A commander who signs a product owns it. The tool that drafted it does not.
 - e. The foundational soldier and staff skills that Army must retain as a capability in their own right, because on operations the tools will be degraded, denied or deceived.
15. A simple rule for autonomy follows.

Consequence × Reversibility × Verifiability

16. Greater AI autonomy is acceptable where the consequence of error is low, the action can be reversed, and the accountable person can readily check the output. Greater human control and assurance are required where consequence is high, effects are irreversible, or the output cannot be verified. Where the tool may be unavailable on operations, the person retains the skill regardless. This rule is deliberately the same shape as the decision rule in the training approach. Leaders should reason the same way about both.

4. What This Means for Army People

17. AI competence is becoming part of military competence, not a specialist skill beside it. Prompt technique is a small part of it. Competence is the ability to employ AI against real Army tasks with judgement, within the security and accountability those tasks require.
18. Four levels, cumulative:
 - a. **All ranks.** Understand what these tools are, how they fail, what may and may not be put into them, and the obligation to verify before use.
 - b. **Practitioners: junior leaders, staff officers and instructors.** Frame a task so a tool can help; evaluate and verify output critically; integrate tools into real workflows such as orders, planning and training design; know when not to use them.
 - c. **Commanders and leaders.** Set risk appetite by consequence; assure outputs; protect the underlying competence of their people; own accountability for AI-enabled products; understand what a system cannot tell them.
 - d. **Specialists.** Data, security and model assurance, and the integration of approved tools into Army systems. Few, and shared across the NZDF.

19. Army's own experience is consistent on one point: confidence and competence come from practice against real tasks, not from exposure. Investment in tools without practised people has not produced productivity in the wider economy, and will not produce capability in Army.

5. What This Means for Individual Training

20. The training approach holds. AI is an enabler of it, not a change to it. Tested against the approach:
- a. **Operational competence.** For many roles the standard now includes the competent, judged use of approved tools. For foundational and safety-critical skills the standard is unchanged and is assured without the tool. Where a skill must survive denial, it is assessed unaided.
 - b. **Common standard, adaptive pathway.** AI makes adaptive pathways practical. Learning data, individual practice and feedback at scale, and assessment support let schools vary the route without varying the standard. This is the largest single contribution AI can make to the training system.
 - c. **Progression and the decision rule.** Tools are introduced where they will be used, not at the foundation. Core skills are built without them; tools enter at guided and independent practice for the tasks that will employ them; performance under pressure is assessed with them and, where it matters, without them. The decision rule sets which.
 - d. **Continuous development.** AI supports learning before, during and after the course: preparation, reinforcement, and rapid update of content when tactics change. It shortens the cycle from feedback to reinforcement and adaptation.
 - e. **The instructor.** Instructors shift from transmitting knowledge to coaching, verifying and assuring. They must be competent with the tools before their students are, and they are the guard against dependency. Instructor development is the first investment.
 - f. **What remains human.** Judgement under pressure, leadership, physical and field skills, the ethical decision, and the trust between people on which teams are built.
 - g. **Preventing dependency.** Deliberate tool-denied training and assessment for foundational skills, with the same discipline Army applies to navigation without GPS. Dependency is prevented by design, not by prohibition.

6. Key Risks and Constraints

21. The risks that matter most for Army, in order of consequence:
- a. **Plausible error.** Fabricated facts, references and figures in products that read as authoritative. Mitigated by verification as a duty, and by never using unverified output where consequence is high.
 - b. **Security and classification.** Sensitive information entering tools not approved for it; sovereignty over where models run; allied caveats. The constraint that most limits early use, and rightly.
 - c. **Loss of competence.** Staff and soldiers who cannot perform the task unaided when the tool is denied. Mitigated through the training approach, not through policy alone.
 - d. **Adversarial manipulation.** Poisoned data, injected instructions and deception aimed at AI-enabled processes. Poorly understood, and growing with dependence.
 - e. **Bias and trust,** especially in decisions about people. Requires a human decision that can be explained.
 - f. **Accountability drift.** Decisions made because the system said so. Mitigated by the autonomy rule and by command practice.
22. The constraints are institutional before they are technical. Army has basic tools on restricted networks; external resources are blocked; security approval is slow; data sits in legacy systems; no one clearly owns AI; procurement runs in years while the capability changes in weeks; the workforce has no spare capacity to experiment; and risk appetite defaults to no. Technical availability is not the barrier. Army's ability to exploit what is available is.

7. An Army Approach to Employing AI

23. The following is proposed for assessing any AI application. It improves the starting model in two ways. It inserts the constraint on the task before any opportunity is considered, so that reasoning begins with what actually limits the output rather than with the tool. And it closes as a loop on measured effect, so that applications are scaled, adjusted or stopped on evidence.

Output → Task → Constraint → AI Mechanism → Human Role → Consequence and Assurance → Competence and Practice → Effect

24. In use:
- Output.** What must Army deliver, and to what standard?
 - Task.** Which task limits that output?
 - Constraint.** What limits the task: time, quality, scale, understanding or people?
 - AI mechanism.** Would AI relieve that constraint, and how? If the mechanism cannot be stated, stop.
 - Human role.** Augment, automate or enable; and what stays with the person.
 - Consequence and assurance.** Apply Consequence × Reversibility × Verifiability. Set the control and verification required.
 - Competence and practice.** Who must be able to do what, and how will they practise it?
 - Effect.** Measure the effect on the output, not the use of the tool. Scale, adjust or stop.
25. A commander, staff officer or training lead should be able to run this on a single page. An application that cannot survive steps c and d is novelty.

8. Principles

26. Seven principles for responsible and effective employment:
- Effect is the test.** AI is adopted where it improves an Army output, and is measured against that output.
 - Reason from the task to the tool,** never the reverse.
 - The commander decides and owns.** AI advises, drafts and analyses. Where consequence is high it does not decide, and accountability does not move.
 - Autonomy proportionate to consequence,** reversibility and verifiability.
 - Competence before dependence.** No one relies on a tool for a skill they must be able to perform without it when it matters.
 - Verify what you use. You own what you sign.**
 - Practise, measure, adapt.** The capability changes in weeks. Army's ability to exploit it is a practised skill, not a purchase.

9. Priority Actions

27. Five actions, in order:
- Own it.** Assign accountability for AI in Army and set a risk appetite by consequence tier, so that low-consequence use on approved tools is enabled by default rather than blocked by default.
 - Prove effect in three places.** Bounded, measured trials: planning and orders production in a headquarters or school; training design, content adaptation and feedback in Army Training Group; and readiness data integration. Measure effect on the output. Stop what does not work.
 - Train it through the approach.** Tiered AI competence built into individual training against real tasks, with verification and tool-denied assurance of core skills. Instructors first.
 - Fix the data.** Readiness and training data are the constraint on the highest-value application Army fully controls.

- e. **Compress the lessons cycle.** Use AI to move from observation to updated training and tactics at a pace the force can act on, and hold the organisation to that pace.

10. Unresolved Questions

28. The following remain open, and the analysis above should be read with them in mind:
- a. Where Army's models and data may run, under whose sovereignty and classification, including on deployed and coalition networks.
 - b. How operational effect, rather than tool use or hours saved, will be measured.
 - c. New Zealand's policy position on AI in targeting and lethal decision, and its coalition obligations.
 - d. How loss of underlying competence will be detected before it matters.
 - e. Who does this work in an army of Army's size, and what it displaces.
 - f. Whether procurement and security approval can move at the pace the capability changes.
 - g. What the AI competence standard is, by rank and trade, and who sets it.

What Should Army Be Able to Do Better Because AI Exists?

Army should be able to plan faster with more options genuinely considered; see its own readiness and its environment sooner and more completely; generate trained people more adaptively, along different routes to the same standard; learn from experience and change its training and tactics at a pace the force can act on; and return the time of its most experienced people to judgement, leadership and training. It should be able to do all of this while remaining able to fight when the tools are degraded, denied or deceived. A small army that cannot do the latter has not gained effectiveness. It has moved its dependence.